ACADEMIC CATALOG

CS 549 Distributed Systems and Cloud Computing

The objective of this course is to give students a basic grounding in designing and implementing distributed and cloud systems, including issues in the implementation of backend services in the cloud itself. What are global consensus and Paxos, and what is their application in building cloud systems? What are the advantages and disadvantages of using distributed NoSQL stores such as Cassandra instead of relational stores such as MySQL? What are strong and weak consistency, what are the "CAP Theorem" and the "CALM Theorem," and what are their implications for building highly available services? What is a blockchain, such as the Bitcoin blockchain, and how does it relate to issues in coordinating distributed systems? What are the roles of REST, Websockets and stream processing in cloud applications? This course will combine hands-on experience in developing cloud services, with a firm grounding in the tools and principles for building distributed and cloud applications, including advanced architectures such as peer-to-peer and publish-subscribe. Besides cloud services, we will also be looking at cloud support for batch processing, such as the Hadoop framework, and its use with NoSQL data stores, such as Cassandra. Prerequisites: Graduate students - Undergraduate data structures and algorithms OR CS 590. Undergraduate students - CS 385.

CREDITS

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PREREQUISITE

(CS 590 or CS 385) and Graduate Student or At Least a Junior

DISTRIBUTION

Computer Science Program

TYPICALLY OFFERED PERIODS

Fall Semester Summer Semester Summer Session 1